

Angelica Knudsen

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EDUCATION

Massachusetts Institute of Technology (MIT)

Candidate for Bachelor of Science, Mechanical Engineering with Energy

Cumulative GPA: 4.6/5.0

Cambridge, MA

May 2026

EXPERIENCE

Geo@MIT Researcher - Geothermal Heat Exchanger System Design

MIT D-Lab

Cambridge, MA

Sep 2025 – Present

- Designing a pilot-scale heat exchanger system to investigate potential changes in water quality from connecting heat exchangers to water mains
- In partnership with the Cambridge Water Department

MIT Researcher - Engineering for Bird Conservation

MIT D-Lab

Cambridge, MA

Feb 2026 – Present

- Co-designing with partners in the Peregrine Fund a fumigation pole to treat the critically endangered Ridgway's hawk from botfly infestations
- Working to increase the height of pole from 24 to 50+ feet, as well as enabling wireless remote operation of the pump and camera

MIT Wind Build Lead

MIT Edgerton Center

Cambridge, MA

Feb 2025 – May 2026

- Led the mechanical and electrical subteams on the design of a small-scale on-shore wind turbine for the 2026 Collegiate Wind Competition hosted by the US DoE

Hatsopoulos Microfluids Laboratory Researcher

MIT Department of Mechanical Engineering

Cambridge, MA

Sep 2024 – May 2025

- Measured static and roll-off contact angle of water droplets on hummingbird feathers using a contact angle goniometer
- Examined feather samples' hierarchical structure via scanning electron microscopy
- Fabricated synthetic feathers with UpNano two-photon polymerization 3D-printer to evaluate and compare mechanical and optical properties with natural hummingbird feathers

PROJECTS

Manta and Remora - Class Project for Engineering Systems Design

Sep 2025 – Dec 2025

- Co-designed a seamlessly coupled remotely-operated vehicle (ROV) and unmanned surface vehicle (USV) to increase the cost-effectiveness, time-efficiency, and sustainability of offshore wind farm foundation inspections
- Contributed most to the custom reel mechanism, designed to withstand over 1000N of force and hold 300m of tether for the ROV

2025 Collegiate Wind Competition Scaled Wind Turbine - Build for MIT Wind

Jan 2025 – May 2024

- Co-designed a small scale offshore wind turbine as part of the US DoEs annual Collegiate Wind Competition
- Specialized in designing a floating foundation inspired by the spar-buoy, tension-leg platform, and barge designs

ADDITIONAL

Languages: Python, MATLAB, C++, C, LaTeX, HTML/CSS

Manufacturing: Laser Cutter, Mill, Bandsaw, Drill Press, Lathe, 3D printing, Soldering

CAD: SOLIDWORKS, Autodesk Fusion 360, Rhino, Onshape, AutoCAD, Revit